

DA6360 - WS.5

1.  $x \in \mathbb{R}^3$       $x = (x_1, x_2, x_3)$

$$\|x\|_{\infty} \leq 5$$

$$|x_i| \leq 5 \quad i = 1, 2, 3$$

$$\|x\|_{\infty} = \max(|x_1|, |x_2|, |x_3|)$$

$$\|x\|_1 = |x_1| + |x_2| + |x_3|$$

$$\|x\|_1 \leq 15$$

$$x = (5, 5, 5)$$

$$\|x\|_{\infty} = 5$$

$$\|x\|_1 = 15$$

$$C = 15$$

$$\boxed{\|x\|_1 \leq n \|x\|_{\infty}}$$

2.  $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ,  $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$B_{\infty}(x_0, 3) \cap B_2(x_1, 3)$$

$$B_{\infty}(x_0, 3) = \{x : \|x - x_0\|_{\infty} < 3\}$$

$$\|x - x_0\|_{\infty} = \max(|x_1 - 1|, |x_2 - 1|)$$

$$|x_1 - 1| < 3, \quad |x_2 - 1| < 3$$

$$-2 < x_1 < 4$$

$$-2 < x_2 < 4$$

$$B_2(x_0, 3) = \{x : \|x - x_0\|_2 < 3\}$$

$$\|x - x_0\|_2 = \sqrt{(x_1 - 1)^2 + (x_2 - 0)^2}$$

$$(x_1 - 1)^2 + x_2^2 < 9$$

circle at  $(1, 0)$ , rad. 3

3.  $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  in std. basis

$$v = \left\{ v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$x = av_1 + bv_2$$

$$a \begin{bmatrix} 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2a + b \\ a + 2b \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$a = \frac{1}{3}, \quad b = \frac{1}{3}$$

$$x_v = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$x_v = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$$

$$4. \quad T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$M_T = \begin{bmatrix} | & | & | \\ e_1 & e_2 & e_3 \\ | & | & | \end{bmatrix}$$

$$T(e_1) = e_2$$

$$T(e_2) = e_3$$

$$T(e_3) = e_1$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$M_T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M_T^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$5. \quad H = \begin{bmatrix} 0.5 & 0.5 \\ 1.5 & 0.5 \end{bmatrix}$$

$$H = VDV^T$$

$$\det(H - \lambda I) = 0$$

$$\det \begin{pmatrix} 0.5 - \lambda & 1.5 \\ 1.5 & 0.5 - \lambda \end{pmatrix} = 0$$

$$(0.5 - \lambda)^2 - (1.5)^2 = 0$$

$$(0.5 - \lambda)^2 = 2.25$$

$$\lambda_1 = 2, \lambda_2 = -1 \quad \text{eigenvalues}$$

$$(H - 2I)u = 0$$

$$\lambda_1 = 2 \quad \begin{bmatrix} -1.5 & 1.5 \\ 1.5 & -1.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$ev_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1 \quad \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$ev_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$



for a sym. matrix to be +ve definite  
all the eigenvalues must be +ve.

2, -1  $\Rightarrow$  not positive definite.

6. (i)  $A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

evs = 1, 1, 1 > 0

symm.

(ii)  $A_2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

sym. diag.

all diag. value > 0

(iii)  $A_3 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

sym.

tridiagonal matrix

7.  $f_1(x) = \frac{1}{2}(5x_1^2 + x_2^2)$

$$f_2(x) = \frac{1}{2}(5x_1^2 - x_2^2)$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(a)  $f(1, 1) = \frac{1}{2}(5+1) = 3$

$$\nabla f_1(x) = \begin{bmatrix} 5x_1 \\ x_2 \end{bmatrix} \Rightarrow \nabla f_1(1, 1) = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\nabla^2 f_1(x) = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T_1(x) = 3 + \begin{bmatrix} 5 & 1 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

$$T_1(x) = 5x_1 + x_2 - 3$$

$$T_2(x) = f_1(x)$$

$$f_2(x) = \frac{1}{2} (5x_1^2 - x_2^2)$$

$$f_2(1,1) = \frac{1}{2} (5-1) = 2$$

$$\nabla f_2(x) = \begin{bmatrix} 5x_1 \\ -x_2 \end{bmatrix} \Rightarrow \nabla f_2(1,1) = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

$$\nabla^2 f_2(x) = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

$$T_1(x) = 5x_1 - x_2 - 2$$

$$T_2(x) = f_2(x)$$

$$(b) \hat{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \text{dec. } f_1$$

a direc<sup>n</sup> dec.  $f_1$  if the directional derivative is  $-ve$ .

$$\nabla f_1(\hat{x})^T d < 0$$

$$x = (2, 2) \quad \nabla f_1(2, 2) = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$$



$$\begin{bmatrix} 10 & 2 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} < 0$$

$$5d_1 + d_2 < 0$$

$$(c) \quad \hat{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{inc. } f_2$$

$$\nabla f_2(\hat{x})^T d > 0$$

$$\nabla f_2(1, 2) = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -2 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} > 0$$

$$5d_1 - 2d_2 > 0$$